



Malai Arasu Paneerselvam

AI Native Technologist | Cloud | AI/ML Native | Distributed Systems

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Professional Summary

Engineering Leader and Technical Manager with 15+ years of experience building scalable, distributed systems and AI-driven platforms across fintech, payments, and commerce domains. I currently serve as an Assistant Vice President of Engineering, where I lead high-performing teams in designing and delivering secure, AI and cloud-native platforms.

I bring a unique combination of hands-on AI engineering, systems thinking, and technical leadership. I have led the design and delivery of multi-tenant AI SaaS platforms, DevOps intelligence systems, AI agents, and LLM-driven analytics solutions. My strength is transforming AI from experimentation into dependable, secure, and measurable business capability—grounded in strong observability, governance, scalability, and ROI.

Technical Skills

AI/ML: Agentic AI, Generative AI, LangGraph, LangChain, MCP tools

Cloud & DevOps: AWS (SageMaker, Bedrock, Lambda, ECS, EKS, S3, Glue, DynamoDB, API Gateway, SQS, Cloudwatch), Terraform, CI/CD, Jenkins, Octopus

Architecture: Microservices, Event-Driven Architecture, Serverless, API-First Design, Distributed Systems

Programming: Java, Python, Spring Boot, Kafka

Databases: Oracle, PostgreSQL, MongoDB, DynamoDB, Opensearch

Tools: GitHub, Jira, Dynatrace Splunk, PagerDuty

Deployment: Docker, Kubernetes

Observability : Dynatrace, Grafana

Certifications

AWS Certified Generative AI Developer - Professional

AWS Certified Solutions Architect – Professional

AWS Certified Machine Learning Engineer – Associate

AWS Certified Developer – Associate

Cloudera Certified Developer (Hadoop)
Oracle Certified Java Programmer

Professional Experience

LPL Financial

Assistant Vice President, Software Engineering
May 2023 – Present | United States

Leading multiple engineering teams for Developer Experience domain, championed a “platform as a product” mindset, treating internal developer platforms with product rigor roadmaps, user feedback loops, and continuous improvement cycles.

- Led the architecture and development multiple next-generation developer platforms enabling agentic AI systems, empowering teams to build autonomous, goal-driven applications using LLMs within a standardized AWS-native ecosystem.
- Designed and delivered reusable frameworks and SDKs for multi-agent orchestration, enabling developers to compose, coordinate, and scale intelligent agents for workflows such as user story creation, development lifecycles
- Developed agent orchestration layers integrating tools, memory, vector stores (S3 vector store and OpenSearch), and external APIs to enable context-aware and stateful AI systems.
- Architected end-to-end pipelines for RAG and agent-driven decision systems, integrating data ingestion, embedding pipelines, retrieval services, and inference orchestration.
- Integrated observability for agentic systems, enabling deep visibility into agent reasoning paths, tool usage, latency, hallucination rates, and execution outcomes using CloudWatch and Dynatrace
- Designed and implemented reusable, scalable frameworks for microservices, APIs, and event-driven architectures, reducing development effort and improving consistency across projects.
- Drove organization-wide initiatives to measure and improve developer productivity using DORA metrics, developer satisfaction (DevEx) KPIs, and engineering analytics dashboards.

Impact: Delivered scalable, secure agentic tools and frameworks that handled high-volume user traffic while enhancing customer experience, resulting in approximately \$2M in cost savings. Improved agent reliability and decision accuracy by implementing evaluation pipelines, guardrails, and observability frameworks.

PayPal

Member of Technical Staff

Jan 2022 – May 2023

Drove the architecture and development of critical platforms within Venmo and SMB onboarding systems.

- Led the end-to-end system design for Venmo Teen Card onboarding, enabling secure, parent-managed financial accounts
- Engineered a scalable onboarding platform supporting millions of users with strong compliance and security controls
- Played a key role in the SMB Reboot initiative, reducing onboarding friction through optimized, minimal-data workflows
- Designed APIs and orchestration flow to improve system performance and user experience

Impact: Reduced teen card onboarding time by 25% by implementing risk-based evaluation strategies, including adaptive verification, real-time fraud scoring, and streamlined compliance checks. Improved onboarding efficiency and scalability while maintaining regulatory compliance in a high-risk fintech environment.

7-Eleven

Senior Software Engineer

Feb 2021 – Jan 2022

Core contributor to the 7Now real-time inventory platform, powering digital commerce experiences.

- Designed and built a real-time inventory and pricing system using event-driven architecture
- Implemented low-latency data processing pipelines supporting millions of transactions
- Contributed to architecture decisions ensuring high availability and fault tolerance

Impact: Enabled accurate, real-time inventory visibility across stores, significantly enhancing digital ordering experience.

Virtusa

Associate Architect

Jan 2018 – Feb 2021

Led architecture design for large-scale financial and payments platforms.

- Architected a Fraud Risk Assessment (FRA) system providing real-time risk scoring for global operations (Citibank)
- Contributed to building a cloud-native mobile commerce platform for enterprise retail clients (PayPal engagement)
- Designed high-throughput, low-latency systems integrating multiple data sources and services

Cognizant

Senior Associate – Technology

Jun 2016 – Jan 2018

Worked on modernizing financial systems for enterprise clients.

- Led cloud migration and modernization of Basel Securitization platform (JP Morgan)
- Improved system scalability and reliability for compliance and regulatory reporting workloads
- Introduced modern cloud and microservices practices to legacy systems

Wipro

Senior Software Engineer

Dec 2009 – May 2016

Built foundational experience in large-scale enterprise systems across commerce and data platforms.

- Contributed to Nike's Commerce in Cloud initiative, building next-gen microservices-based commerce systems
- Developed online usage and analytics platforms for Dun & Bradstreet
- Gained deep expertise in Java-based backend systems, APIs, and distributed architectures

Education

University: Anna University

Degree: Bachelor of Engineering

Field of Study: Computer Science

Year: 2004 to 2008